

CS 340 Project Two – Grazioso Salvare Dashboard

About the Project

This project expands the CRUD Python module created in Project One into a fully functional web dashboard for Grazioso Salvare. The dashboard connects to a MongoDB database containing Austin Animal Center records and allows users to filter shelter animals that meet search-and-rescue training criteria.

The dashboard provides an interactive interface where users can:

- View all shelter animals in an interactive data table
- Filter dogs by rescue type
- View an updated bar chart based on the selected filter
- Display the selected animal's location on an interactive map
- Reset the dashboard to display all records

Required Functionality

The dashboard includes the following required features:

- Grazioso Salvare logo
- Dashboard identifier containing my name
- Interactive filter options
- Interactive data table
- Geolocation map
- Dynamic chart
- Reset option to remove filters

Dashboard Screenshots

Water Rescue Filter



CS-340 Grazioso Salvare Dashboard

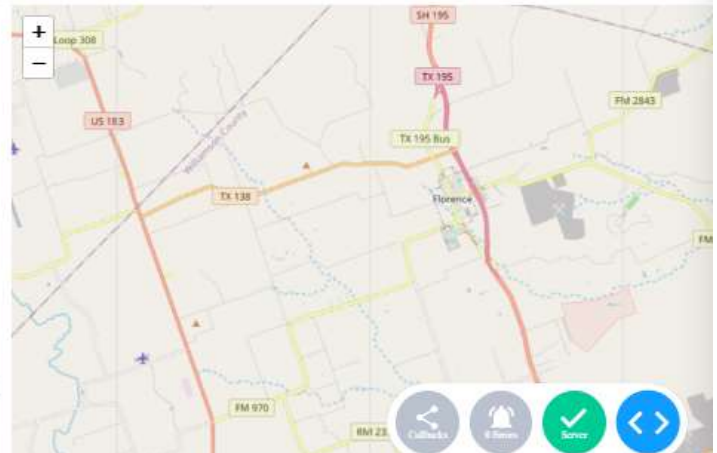
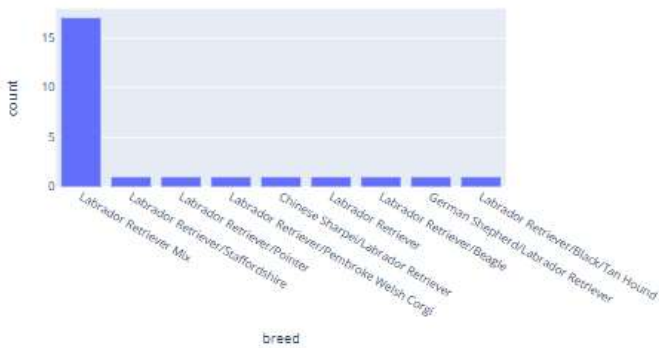
Jennifer J.

☒ Water Rescue ☐ Mountain or Wilderness Rescue ☐ Disaster or Individual Tracking ☐ Reset

rec_num	age_upon_outcome	animal_id	animal_type	breed	color	date_of_birth	datetime	monthyear	name	outcome_subtype	outcome_type	sex_
filter data:												
☒ 36	6 months	A786953	Dog	Labrador Retriever Mix	Yellow	2014-12-06	2015-07-06 11:33:00	2015-07-06T11:33:00		Medical	Euthanasia	Intac
☐ 289	9 months	A669177	Dog	Labrador Retriever/Staffordshire	Black/White	2013-03-16	2013-12-24 14:43:00	2013-12-24T14:43:00	Ally		Adoption	Intac
☐ 732	2 years	A749782	Dog	Labrador Retriever Mix	Tan/White	2015-05-19	2017-07-25 14:59:00	2017-07-25T14:59:00	*Catalina		Return to Owner	Intac
☐ 845	1 year	A672885	Dog	Labrador Retriever/Pointer	Chocolate/White	2013-02-02	2014-02-07 15:00:00	2014-02-07T15:00:00	Hazel	Partner	Transfer	Intac
☐ 986	1 year	A715357	Dog	Labrador Retriever/Pembroke Welsh Corgi	Black	2014-11-04	2015-11-08 00:00:00	2015-11-08T00:00:00	*Roxy	Partner	Transfer	Intac
☐ 1121	1 year	A757158	Dog	Labrador Retriever Mix	White/Black	2016-08-30	2017-08-31 14:12:00	2017-08-31T14:12:00	Pirata		Return to Owner	Intac
☐ 1628	9 months	A748471	Dog	Labrador Retriever Mix	Tan/White	2016-03-17	2016-12-23 17:13:00	2016-12-23T17:13:00	Mika		Adoption	Intac
☐ 1757	7 months	A742767	Dog	Labrador Retriever Mix	Black	2016-06-27	2017-02-14 15:20:00	2017-02-14T15:20:00	Marley		Return to Owner	Intac
☐ 1988	1 year	A762781	Dog	Labrador Retriever Mix	Black/White	2016-11-27	2017-12-03 13:09:00	2017-12-03T13:09:00		Partner	Transfer	Intac
☐ 2841	2 years	A782745	Dog	Labrador Retriever Mix	Black	2013-05-22	2015-05-22 11:45:00	2015-05-22T11:45:00	Abigail		Return to Owner	Intac

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Top 10 Breeds



Mountain or Wilderness Rescue Filter



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Water Rescue

Mountain or Wilderness Rescue

Disaster or Individual Tracking

Reset

rec_num	age_upon_outcome	animal_id	animal_type	breed	color	date_of_birth	datetime	monthyear	name	outcome_subtype	outcome_type	sex_upon_out
filter data.												
1233	7 months	A722362	Dog	German Shepherd/Labrador Retriever	Tan/White	2015-08-25	2016-03-25 12:11:00	2016-03-25T12:11:00	Brisket		Return to Owner	Intact Male
2275	1 year	A668767	Dog	German Shepherd Mix	Black/Tan	2012-11-09	2013-12-09 18:29:00	2013-12-09T18:29:00	Vincent		Return to Owner	Intact Male
3100	1 year	A742163	Dog	German Shepherd/Labrador Retriever	Tan	2016-03-17	2018-01-18 17:50:00	2018-01-18T17:50:00	Osito		Return to Owner	Intact Male
3130	2 years	A721834	Dog	Siberian Husky	Brown/White	2014-03-05	2016-03-23 16:23:00	2016-03-23T16:23:00		Suffering	Euthanasia	Intact Male
3244	7 months	A692441	Dog	Siberian Husky Mix	Red/White	2014-04-07	2014-11-20 18:07:00	2014-11-20T18:07:00	Wolfe	Suffering	Euthanasia	Intact Male
3930	10 months	A743413	Dog	Siberian Husky Mix	Black/White	2016-04-11	2017-02-11 13:18:00	2017-02-11T13:18:00	Lobo		Return to Owner	Intact Male
4618	1 year	A699090	Dog	Rottweiler Mix	Black/Brown	2014-03-22	2015-03-23 18:46:00	2015-03-23T18:46:00	Rex		Return to Owner	Intact Male
5032	2 years	A765283	Dog	German Shepherd Mix	Black/Tan	2015-01-16	2017-11-05 17:05:00	2017-11-05T17:05:00	Lu		Adoption	Intact Male
5273	5 months	A738258	Dog	German Shepherd Mix	Black/Brown	2016-01-01	2016-07-01 13:36:00	2016-07-01T13:36:00	Teddy		Return to Owner	Intact Male
5315	2 years	A708726	Dog	Alaskan Malamute	Sable/White	2013-07-30	2015-08-02 17:24:00	2015-08-02T17:24:00	Papa		Return to Owner	Intact Male

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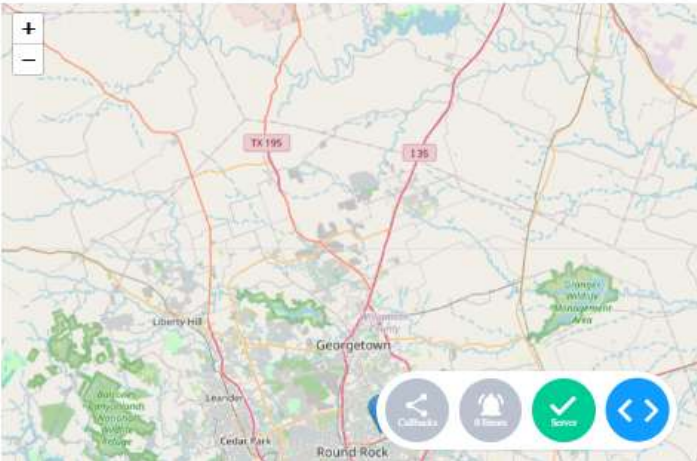
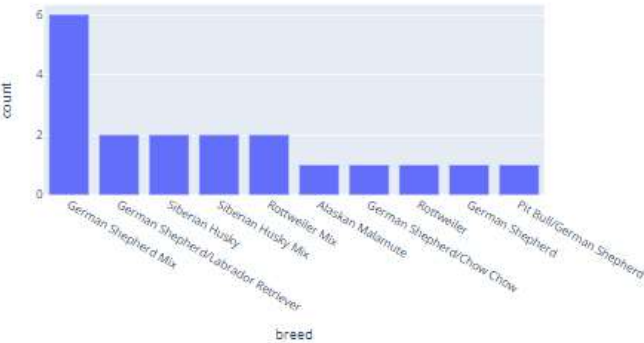
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Top 10 Breeds



Disaster or Individual Tracking Filter



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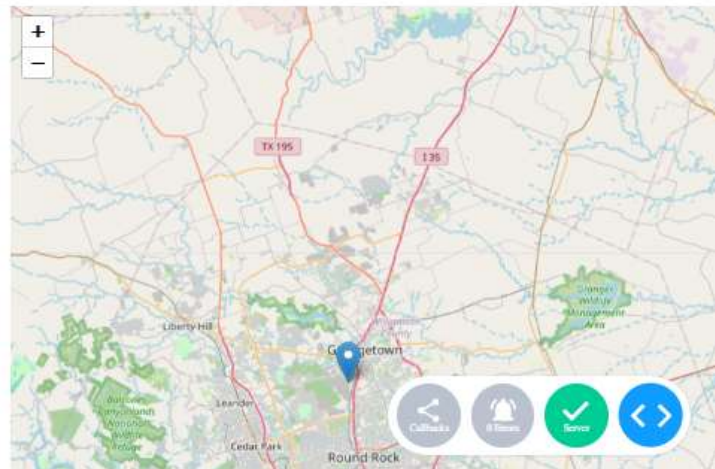
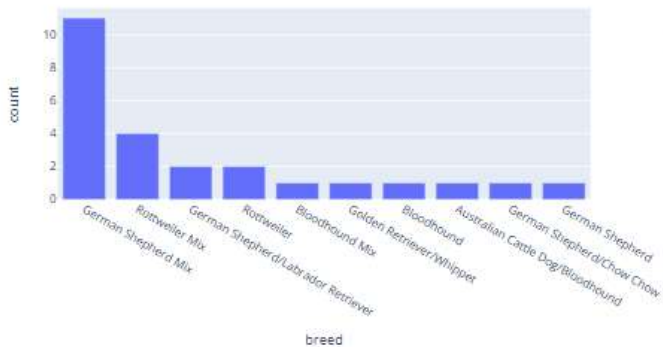
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☐ Water Rescue ☐ Mountain or Wilderness Rescue ☒ Disaster or Individual Tracking ☐ Reset

rec_num	age_upon_outcome	animal_id	animal_type	breed	color	date_of_birth	datetime	monthyear	name	outcome_subtype	outcome_type	sex_upon_out
filter data.												
☒ 256	3 years	A717863	Dog	German Shepherd Mix	Brown	2012-12-16	2015-12-20 00:00:00	2015-12-20T00:00:00	*Lucky	Partner	Transfer	Intact Male
☐ 1233	7 months	A722362	Dog	German Shepherd/Labrador Retriever	Tan/White	2015-08-25	2016-03-25 12:11:00	2016-03-25T12:11:00	Brisket		Return to Owner	Intact Male
☐ 1765	3 years	A757378	Dog	German Shepherd Mix	Tricolor	2014-09-01	2017-10-03 13:28:00	2017-10-03T13:28:00	42	Partner	Transfer	Intact Male
☐ 1918	3 years	A698931	Dog	Bloodhound Mix	Red/Black	2012-03-19	2015-03-20 11:24:00	2015-03-20T11:24:00			Return to Owner	Intact Male
☐ 1977	5 months	A706373	Dog	Golden Retriever/Whippet	Red/Gold	2015-01-13	2015-07-02 11:38:00	2015-07-02T11:38:00	Howie	Partner	Transfer	Intact Male
☐ 2275	1 year	A668767	Dog	German Shepherd Mix	Black/Tan	2012-11-09	2013-12-09 18:29:00	2013-12-09T18:29:00	Vincent		Return to Owner	Intact Male
☐ 2469	3 years	A672744	Dog	Rottweiler Mix	Black/Tan	2011-02-17	2014-04-17 17:09:00	2014-04-17T17:09:00	Oso	Aggressive	Euthanasia	Intact Male
☐ 2987	4 years	A694614	Dog	Rottweiler	Black/Brown	2011-01-01	2015-01-01 14:25:00	2015-01-01T14:25:00	Striker		Return to Owner	Intact Male
☐ 3100	1 year	A742163	Dog	German Shepherd/Labrador Retriever	Tan	2016-03-17	2018-01-18 17:50:00	2018-01-18T17:50:00	Osito		Return to Owner	Intact Male
☐ 3767	4 years	A712291	Dog	Bloodhound	Red	2011-09-20	2015-09-22 15:43:00	2015-09-22T15:43:00	Boomer		Return to Owner	Intact Male

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Top 10 Breeds



Reset Filter



CS-340 Grazioso Salvare Dashboard

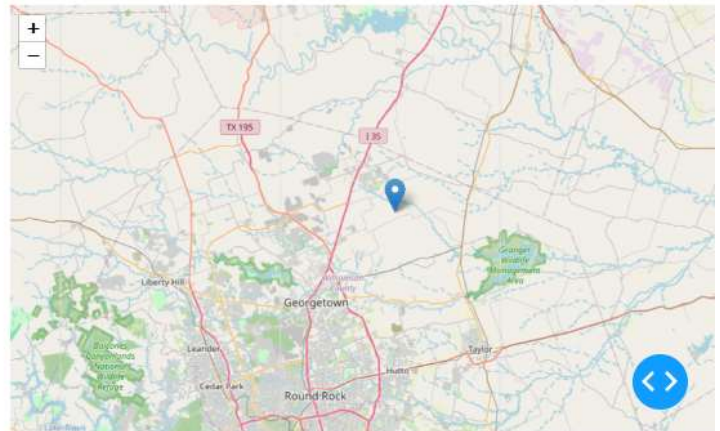
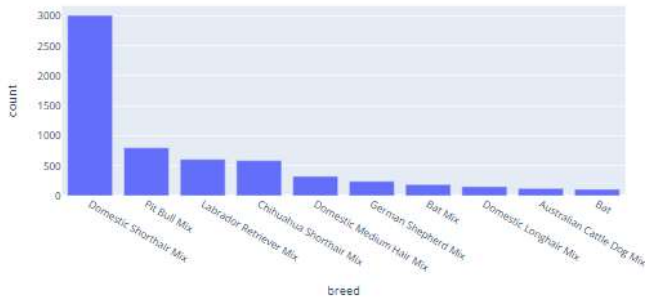
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rec_num	age_upon_outcome	animal_id	animal_type	breed	color	date_of_birth	datetime	monthyear	name	outcome_subtype	outcome_type	sex_upon_outcome	location_lat	location
filter data.														
☒ 3	2 years	A716330	Dog	Chihuahua Shorthair Mix	Brown/White	2013-11-18	2015-12-28 18:43:00	2015-12-28T18:43:00	Frank		Adoption	Neutered Male	30.7595748121648	-97.552375
☐ 9	3 years	A720214	Dog	Labrador Retriever Mix	Red/White	2013-02-04	2016-02-11 12:41:00	2016-02-11T12:41:00	Blessing		Adoption	Spayed Female	30.3870648199411	-97.368433
☐ 10	3 months	A664290	Cat	Domestic Shorthair Mix	Tortie	2013-09-01	2013-12-08 14:58:00	2013-12-08T14:58:00	*Taylor		Adoption	Spayed Female	30.7583105481048	-97.618292
☐ 11	1 year	A721199	Dog	Dachshund Wirehair Mix	Tan/White	2015-02-23	2016-02-27 17:49:00	2016-02-27T17:49:00	Belle		Adoption	Spayed Female	30.7290227261146	-97.375332
☐ 12	1 year	A664843	Dog	Pit Bull Mix	Brown/White	2013-06-09	2014-08-18 17:24:00	2014-08-18T17:24:00	Sherlock	Partner	Transfer	Neutered Male	30.4515549397366	-97.474104
☐ 13	1 year	A700408	Cat	Domestic Shorthair Mix	Brown Tabby/White	2014-04-13	2015-04-15 13:34:00	2015-04-15T13:34:00	Myla		Return to Owner	Spayed Female	30.4181154527976	-97.562415
☐ 14	2 years	A742287	Dog	Boxer/Bulldog Mix	Brown Brindle/White	2015-01-18	2017-02-11 12:30:00	2017-02-11T12:30:00	*Kawhi		Adoption	Neutered Male	30.4551148649096	-97.308778
☐ 15	3 years	A712638	Dog	Pit Bull Mix	Red/White	2012-09-26	2016-07-18 17:52:00	2016-07-18T17:52:00	Marcus	Partner	Transfer	Neutered Male	30.5798299207017	-97.558848
☐ 5	2 years	A691584	Dog	Labrador Retriever Mix	Tan/White	2012-11-06	2015-05-30 13:48:00	2015-05-30T13:48:00	Luke		Return to Owner	Neutered Male	30.7104815618433	-97.562297
☐ 16	5 years	A723742	Dog	Miniature Schnauzer Mix	Black/White	2011-04-05	2016-04-18 17:27:00	2016-04-18T17:27:00	Gretchen		Adoption	Spayed Female	30.4792884863566	-97.408853

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Top 10 Breeds



Tools Used

Python

Python was used as the primary programming language because it integrates well with MongoDB and provides excellent support for data analysis and web development.

MongoDB

MongoDB stores the Austin Animal Center data and serves as the model component of the application.

MongoDB was selected because it:

- Stores data as flexible JSON-like documents
- Integrates directly with Python using PyMongo
- Supports efficient filtering with query documents
- Easily returns data that can be converted into pandas DataFrames

PyMongo

PyMongo provides communication between Python and MongoDB through the CRUD Python module developed in Project One.

Dash

Dash provides the view and controller components of the application. It creates an interactive web interface using Python without requiring JavaScript.

Dash manages:

- Interactive radio buttons
- Data table updates
- Dynamic charts
- Interactive map updates
- Callback functions between dashboard components

pandas

Pandas converts MongoDB query results into DataFrames that can be displayed within the dashboard and visualized using charts.

Plotly Express

Plotly Express creates interactive charts that automatically update when users apply filters.

Dash Leaflet

Dash Leaflet displays the geographic location of the selected shelter animal on an interactive map.

Resources Used

- SNHU CS 340 Module Resources
- Dash Documentation: <https://dash.plotly.com/>
- Plotly Documentation: <https://plotly.com/python/>
- MongoDB Documentation: <https://www.mongodb.com/docs/>
- PyMongo Documentation: <https://pymongo.readthedocs.io/>

Reproducing the Project

To reproduce this project:

1. Install MongoDB.
2. Import the Austin Animal Center dataset into the aac database.
3. Create the aacuser account with appropriate permissions.
4. Place the CRUD Python module and dashboard notebook in the same directory.
5. Add the Grazioso Salvare logo image to the project folder.
6. Install required Python packages, including:
 - pandas
 - dash
 - dash-leaflet
 - plotly
 - pymongo
 - jupyter-dash
7. Update the MongoDB username and password in the notebook.
8. Run all notebook cells.
9. Launch the dashboard.

10. Verify that each filter updates the data table, chart, and map correctly.

Challenges Encountered

One challenge was configuring MongoDB authentication correctly. This was resolved by using the correct MongoDB username, password, and authentication source in the CRUD module.

Another challenge was creating database queries that matched the required rescue categories. This was solved by constructing MongoDB query dictionaries that filtered animals by breed, age, animal type, and sex.

A final challenge was ensuring that all dashboard components updated together. Dash callback functions were used so that the data table, bar chart, and map automatically refreshed whenever a new filter was selected.